

QU ACM STUDENT CHAPTER

WELCOME TO OUR FIRST

BIG GAME DEV PROJECT

PROJECT ACMO A 3D low-poly retro top-down shooter

SELECT STAGE

WHAT WE'LL TALK ABOUT

01 THE IDEA >

02 HOW WE DO THE WORK

03 THE NEEDED SKILLS

04 THE INITIAL TASKS

05 Q&A



SHOOT BUGS. COLLECT BITS. SPELL WORDS.



GAMEPLAY MOCKUP · LEVEL 3 · TARGET WORD "ACM"

A **3D low-poly retro top-down shooter**. Our mascot **ACMO** fires plasma beams from its screen at bugs that carry screens of their own.

0 1	Hit a **bit bug** → its bit is added to your bits bar
P	Hit **Parse** → the bar becomes a character
C	Hit **Clear** → the bar resets

GOAL

Each level gives you a word. Parse its letters correctly to clear the level.

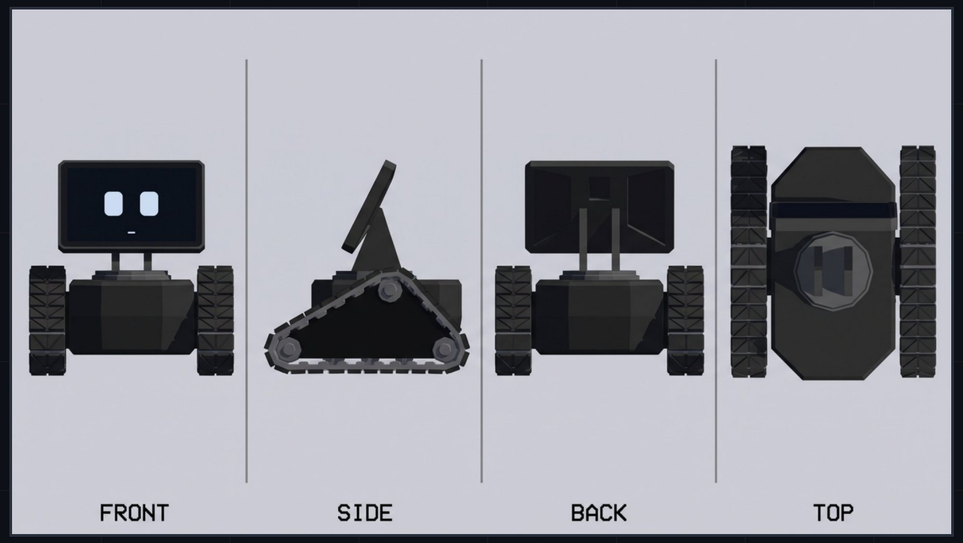
ONE TOUCH = GAME OVER

Your survival time goes on the leaderboard.

MEET ACMO



HERO RENDER



TURNAROUND · FRONT / SIDE / BACK / TOP

01 Based on our real mascot robot: two triangular **tank tracks** and a squat black body

02 The **head is a turret**: tracks drive the body, the screen rotates on its own to aim

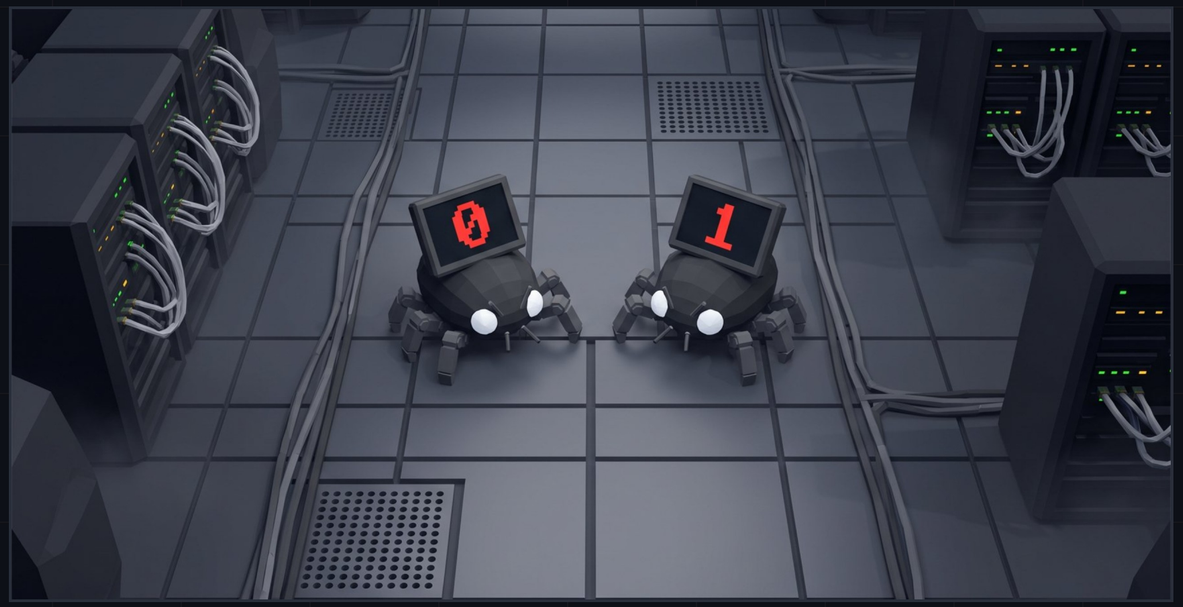
03 The **screen is the face and the weapon**: it fires the plasma beam



ACMO OS · 9 FACES FOR GAME EVENTS

THE BUGS

BIT BUGS 0 1

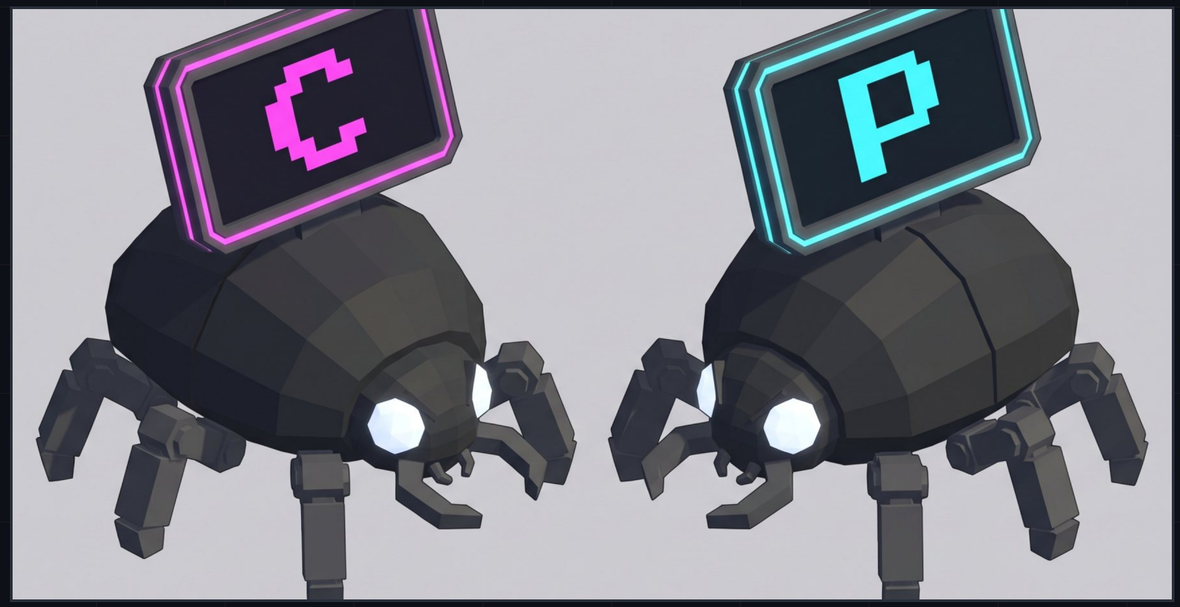


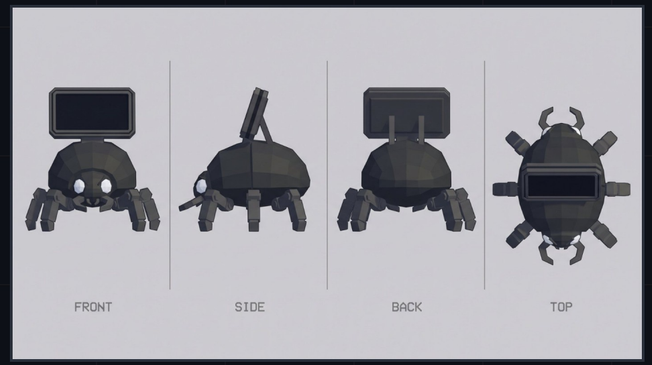


Small, fast mechanical mites. The screen on their back shows a **red 0 or 1**.

Hit one → that bit joins your bits bar.

ACTION BUGS C P





Bigger armored beetles with a glowing frame.

C clears the bar.
P parses it into a letter.

FROM BITS TO LETTERS

BITS:

0	1	0	0	0	0	0	1
128	64	32	16	8	4	2	1

HIT P

>>



64 + 1

= 65

ASCII code of A

LEVEL 1 · TARGET [A][C][M]

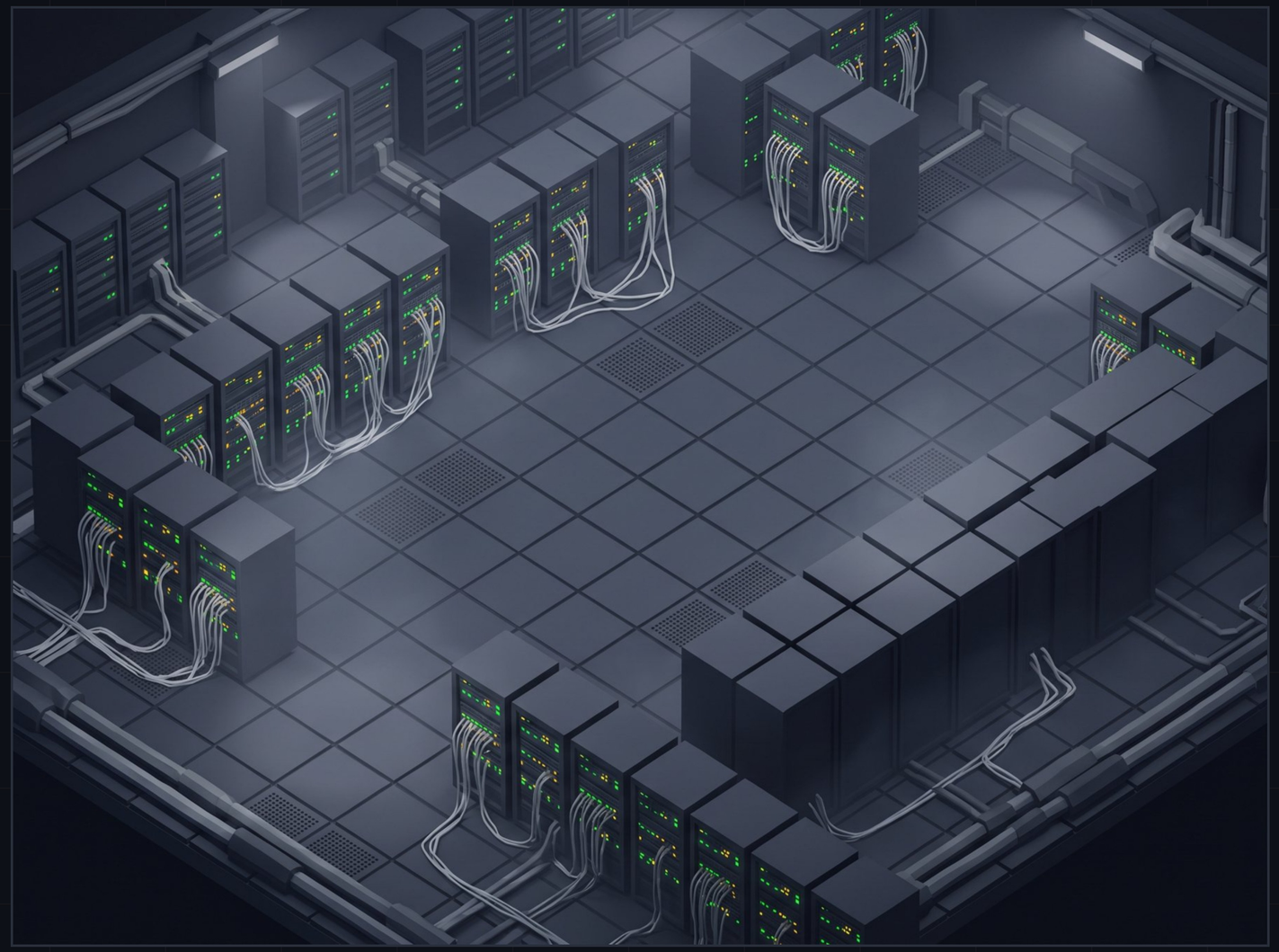
every level = a new word

LETTER	BITS TO COLLECT	DECIMAL
A	01000001	65
C	01000011	67
M	01001101	77



SCREEN GLYPHS · MADE A MISTAKE?
HIT C

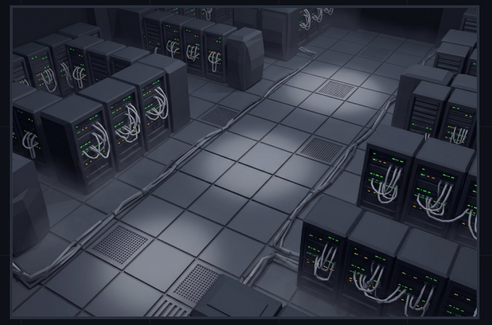
THE WORLD: A DATA CENTER



ARENA MOOD · OPEN FLOOR IN THE MIDDLE FOR COMBAT

Built from a **modular low-poly kit**: floor tiles, rack modules and props snap together on a grid.

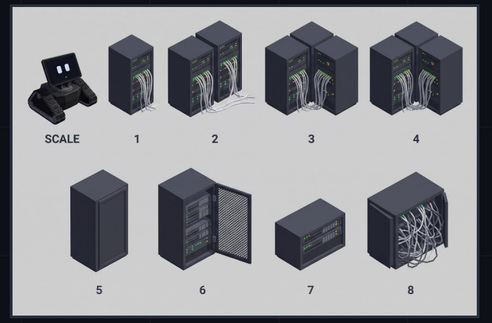
The room only uses green and amber lights, so red, cyan and magenta always mean gameplay.



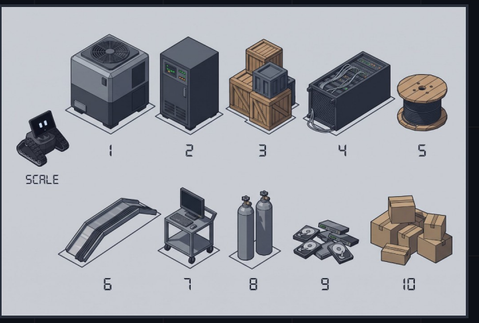
AISLE



8 FLOOR TILES



8 RACK MODULES

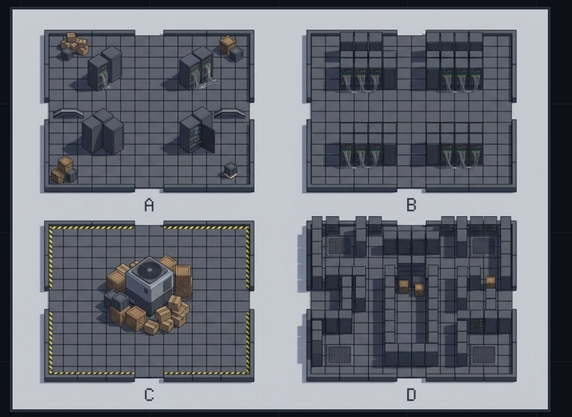


10 PROPS / OBSTACLES

IT NEVER ENDS

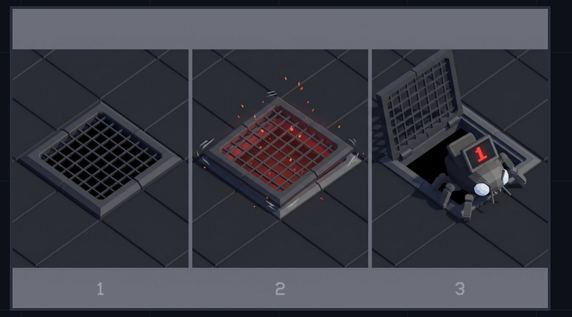


EARLY → MID → LATE · BUGS GET FASTER AND MORE FREQUENT OVER TIME



PROCEDURAL MAP

Map chunks with matching openings, picked and rotated at random.



SPAWN VENTS

A red glow warns you before a bug climbs out.

Words generate forever. When a bug touches you:

GAME OVER > YOUR NAME > LEADERBOARD

Ranked by time survived

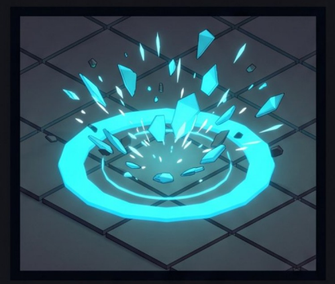
EVERY HIT FEELS GOOD



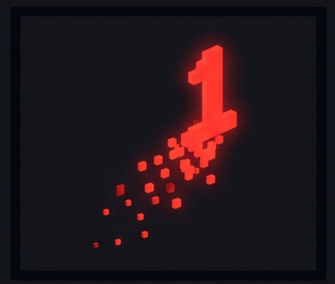
PLASMA MUZZLE FLASH 1



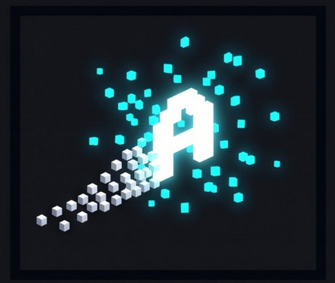
PLASMA BEAM 2



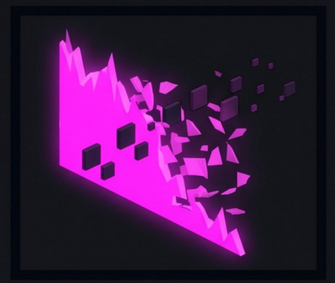
BEAM IMPACT 3



BIT PICKUP 4

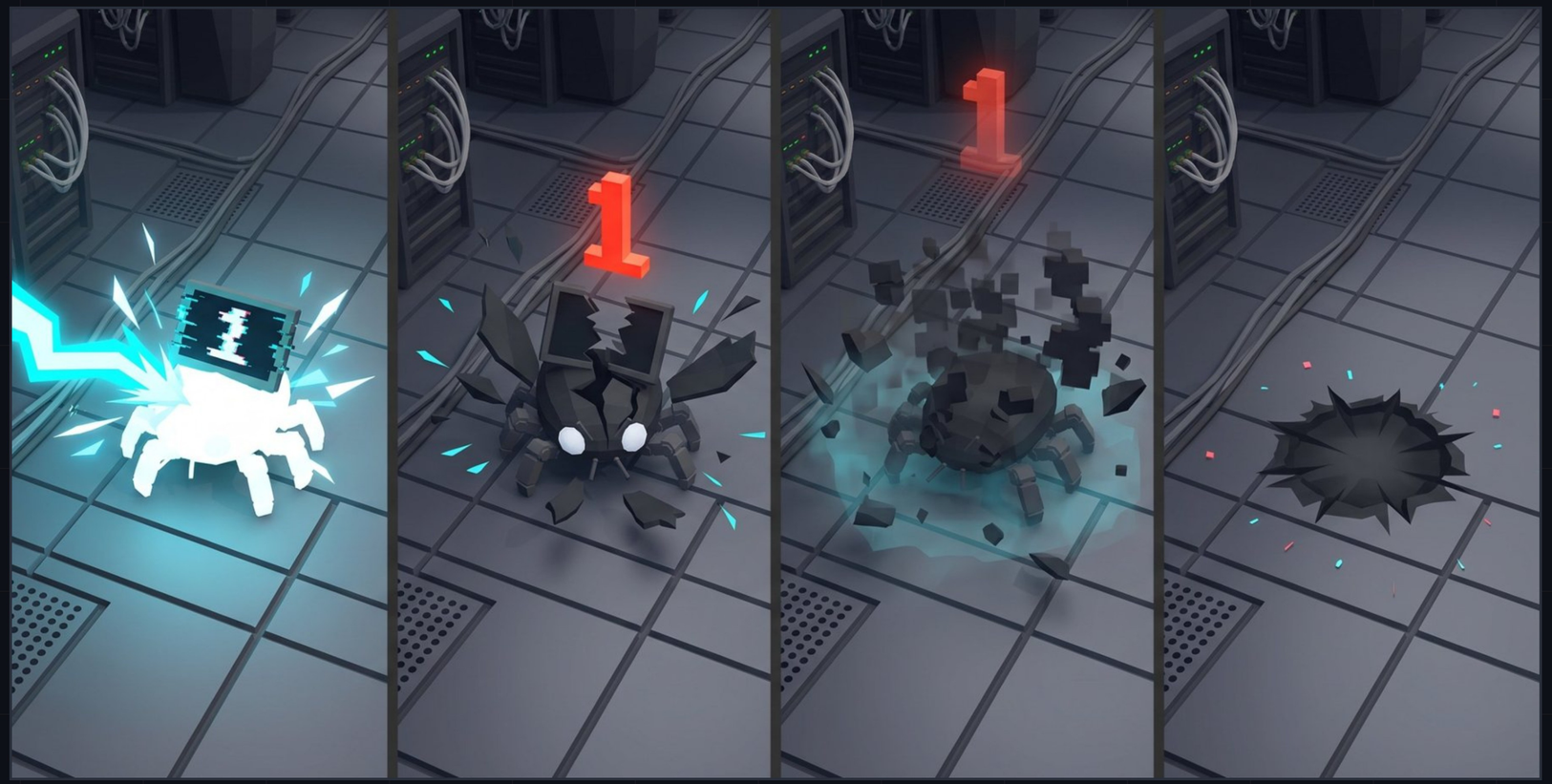


PARSE SUCCESS 5



CLEAR 6

VFX SHEET · 6 EFFECTS



BUG DESTRUCTION · FLASH → CRACK → SCATTER → SCORCH

SHOOT
Muzzle flash + plasma beam

HIT A BIT
The digit flies up to your bar

PARSE / CLEAR
Letter burst or magenta wipe

HOW WE WORK

1 SETUP

A public GitHub repo + a GitHub Project with detailed tasks for everyone

2 PICK A TASK

Select a task and start working on it

3 DRAG TO REVIEW

When it's done, drag the card to the **Review** column

4 REVIEW → DONE

I review it and flag it as **Done**

GITHUB PROJECT · ACMO GAME

EXAMPLE BOARD

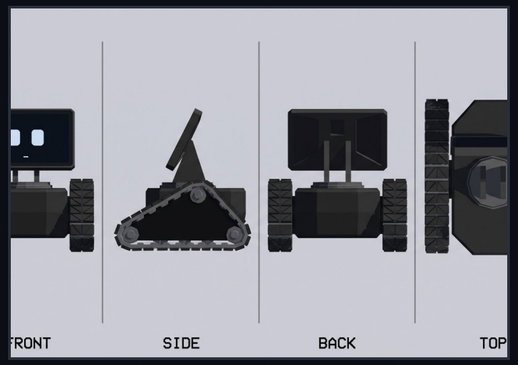
TO DO	IN PROGRESS	REVIEW	DONE
Code ACMO movement C#	Model ACMO 3D @you	drag here when done	flagged after review
Level 1: word "ACM" C#			

Repo: [REPO LINK]

CHOOSE YOUR SKILLS

SKILL 01

3D DESIGN



Low-poly modeling of ACMO, the bugs and the data-center kit, straight from the reference sheets.

SKILL 02

C# PROGRAMMING

```
string bits = "01000001";
char c = (char)Convert
   .ToInt32(bits, 2);
// c == 'A'
```

Movement, shooting, enemies, the bits bar, words and levels.

SKILL 03

USING CODING AGENTS

```
> add an 8-slot bits bar to the HUD
reading project files…
+ BitsBar.cs
run it · review the diff
```

Use AI agents to move faster. You still read, test and understand every line you submit.

SKILL 04

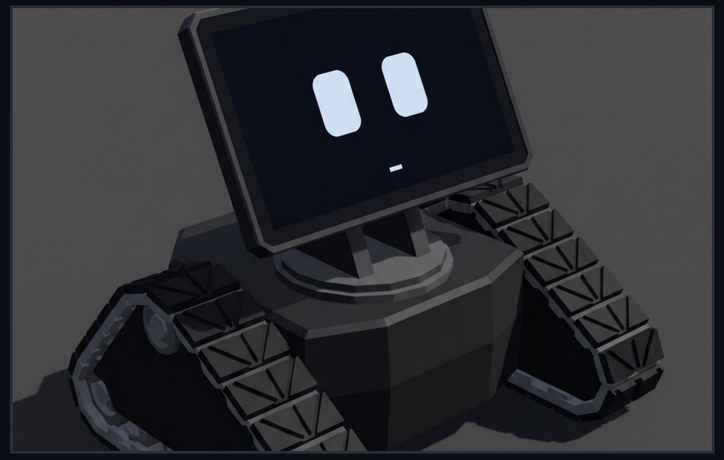
PATIENCE

LOADING GAME…

one task at a time

This is our first big game. Things will break; we debug, learn and keep going.

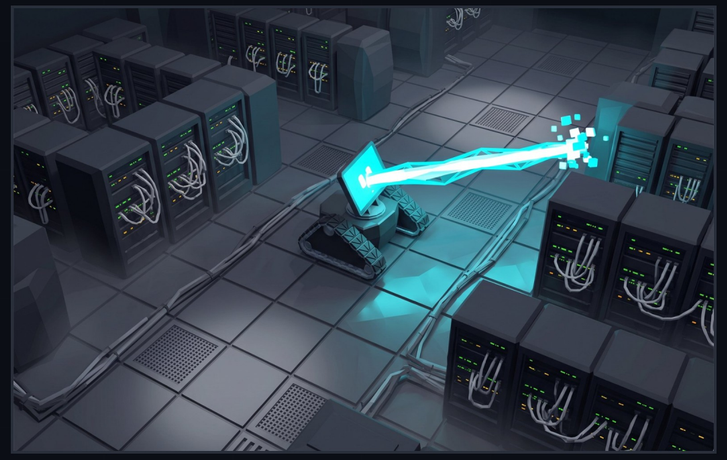
FIRST MISSIONS



TASK 01 — 3D

CREATE THE MODEL FOR THE CHARACTER

Low-poly ACMO with separate parts: two track pods, body, turntable and the screen head.



TASK 02 — C#

CODE THE MOVEMENT OF THE CHARACTER

The tracks drive the body around the arena; the head turns on its own to aim.



TASK 03 — C#

CODE THE FIRST LEVEL: ACM

The first level's word is ACM: collect the bits, parse A, C and M to win.



LEVEL COMPLETE

THANK YOU!

Q$A ASK ANYTHING

gu.acm.org · @acm.gu

Repo: [REPO LINK]